Hires Before Headcount

How the U.S. information sector shrank without firing

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Abstract

The U.S. information sector has lost 381,000 jobs since its November 2022 peak, a decline of 12.2 per cent, and it has stayed below that peak for 47 consecutive months. The usual explanation is a wave of layoffs. The job-openings and labour-turnover data say otherwise: layoffs in the sector averaged 49,000 a month in 2026 against 33,000 before the pandemic, while hires fell from a peak of 140,000 a month in October 2021 to a 2026 average of 80,000 and quits fell 23 per cent below their pre-pandemic average. The sector did not fire its way down. It stopped replacing the people who left.

That timing matters more than the magnitude. Hires peaked in October 2021, thirteen months before employment peaked and thirteen months before the release of ChatGPT. The hiring rate was already falling through 2022, and the first negative year-over-year reading in hires arrives in February 2022. Any account that attributes the sector's contraction to generative AI has to explain why the entry door closed first.

The paper also reports a negative result that cuts against part of the displacement story. If AI were substituting for junior work, the unemployment rate of workers aged 20 to 24 should have risen relative to prime-age workers. It rose more in percentage points (2.0 against 0.7) but the ratio between the two is unchanged: 2.08 now against 2.07 in 2016–2019 and 2.12 in 2023. What looks like an entry-level effect in levels is ordinary cyclical incidence in relative terms. The aggregate cohort data cannot see the composition change the hypothesis predicts.

The contribution is a mechanism test. A substitution story predicts adjustment through the hiring margin before the separation margin; the data show exactly that pattern, and they show it starting a year before the technology existed in public form. The mechanism is real, the attribution is not identified, and the measurement that would settle it is hiring composition by seniority, not another exposure index.

Keywords: labour demand, hiring, separations, JOLTS, generative artificial intelligence, attrition

JEL: J21, J23, J63, O33

1. Introduction

Most arguments about artificial intelligence and employment are arguments about job destruction. A model can perform a task; the task is part of an occupation; therefore the occupation is exposed. The implied mechanism is a separation: a worker is let go because a machine can do the work.

Firms rarely adjust employment that way. The first and cheapest margin is the entry door. A firm that expects to need fewer analysts stops hiring analysts, allows attrition to reduce headcount, and only later considers redundancies, which carry legal cost, severance, morale effects and public attention. A hiring freeze shows up in job openings and hires quarters before it shows up in payrolls. If the AI substitution story is right, that is the order in which the data should move.

This paper tests the order. It uses the U.S. Job Openings and Labor Turnover Survey, which counts hires and separations separately, alongside sector employment and unemployment by age cohort, for January 2016 to October 2026. The focus is the information sector, the sector with the largest employment break since the pandemic and the sector that produces the technology.

Three findings structure the paper.

First, the entry door closed long before the sector shrank. Hires in the information sector peaked in October 2021 at 140,000 a month and have not recovered; the 2026 average is 80,000. Employment peaked thirteen months later, in November 2022. The first negative year-over-year reading in hires is February 2022, nine months before the release of ChatGPT.

Second, the contraction ran through attrition rather than dismissal. Layoffs in 2026 average 49,000 a month, well above the pre-pandemic 33,000 but far below the 191,000 reached in March 2020. Quits fell to 34,000 a month from 44,000 before the pandemic. The hire-to-separation ratio averaged 0.89 in 2026, so the sector lost workers every month without a mass separation event.

Third, the obvious entry-level test fails at the level the data can see. Workers aged 20 to 24 did see unemployment rise two percentage points from its April 2023 low, against 0.7 points for workers aged 25 to 54. But the ratio between the two rates is flat: 2.08 in October 2026, 2.12 in 2023, 2.07 in 2016–2019. The extra rise is the normal cyclical penalty of being young, not evidence of a new seniority-specific shock. A real test needs hiring composition by seniority, which neither JOLTS nor the monthly CPS provides.

1.1 Relation to the companion paper

A companion paper, *Exposure Is Not Adoption*, examines sector employment through September 2026 and concludes that the information-sector break cannot be attributed to generative AI from sector aggregates. This paper goes one level down, to the flows that produce the stock. It does not overturn that conclusion; it sharpens the mechanism and adds a negative result. The companion paper's five predictions, particularly P1 (entry-level composition moves first) and P3 (adoption predicts hiring better than exposure does), are the framework this paper tests against.

1.2 What this paper does not do

It does not identify a causal effect. It does not claim that AI caused the hiring freeze, because the freeze begins before the treatment. It does not use firm-level data, which the public JOLTS does not provide. And it does not test occupation-level composition, because neither data set used here contains occupation by seniority.

2. Two stories, one door

Two explanations for the information sector's contraction are live, and they predict the same flow pattern.

The substitution story. Generative AI reduces the demand for junior cognitive labour. Firms that would have hired analysts, researchers, copywriters and support staff hire fewer. Because firing is costly, the adjustment runs through fewer hires and normal attrition. Employment falls slowly; hires fall first; quits fall as opportunities disappear; layoffs stay low relative to a recession. The prediction is a hiring-led contraction concentrated at entry level.

The cost-of-capital story. Higher interest rates reduce the present value of long-duration projects, and the information sector is the most long-duration sector in the economy. Firms cut discretionary spending, and hiring is discretionary spending. Advertising and digital media, a large part of the sector, contract with the cycle. The prediction is the same: fewer hires, attrition, low layoffs, and a contraction that begins with the rate cycle rather than with any technology.

The two stories differ in timing and in composition, not in flow mechanics. Timing: the substitution story dates the turn to late 2022 or 2023, the rate story to 2021–2022, when the Federal Reserve began signalling tightening. Composition: the substitution story predicts the freeze is concentrated in junior roles; the rate story predicts it is broad, and concentrated in firms and projects that depend on external finance.

The data in this paper can test the timing and the flow mechanism. They cannot test composition. That asymmetry is the paper's central limitation and its main result.

3. Data and method

3.1 Sources

All data are from the U.S. Bureau of Labor Statistics public API, retrieved November 2026, seasonally adjusted.

Table 1. Series used

Series	Definition	Role
JTS510000000000000HIL	Information: hires, thousands	Entry margin
JTS510000000000000TSL	Information: total separations, thousands	Exit margin
JTS510000000000000LDL	Information: layoffs and discharges, thousands	Involuntary exit
JTS510000000000000QUL	Information: quits, thousands	Voluntary exit
JTS510000000000000JOL	Information: job openings, thousands	Posting margin
JTS000000000000000HIL	Total nonfarm hires, thousands	Economy-wide comparison
CES5000000001	Information: employment, thousands	Stock
LNS14000012	Unemployment rate, 16–19, per cent	Entry cohort
LNS14000036	Unemployment rate, 20–24, per cent	Entry cohort
LNS14000060	Unemployment rate, 25–54, per cent	Prime-age comparison

The panel covers January 2016 to October 2026: 130 monthly observations. JOLTS is published with a one-month lag, so the most recent flow observation is September 2026 while the most recent employment observation is October 2026.

3.2 Definitions

The hires rate is hires divided by employment from the establishment survey, expressed as a percentage; it answers how large the entry flow is relative to the stock. Net hires is hires minus total separations; a negative value means the sector lost workers in that month. The hire-to-separation ratio is hires divided by separations; below one, the sector contracts absent reallocation. The youth-to-prime unemployment ratio is the unemployment rate for ages 20–24 divided by the rate for ages 25–54; a rising ratio indicates a shock that falls disproportionately on young workers, a flat ratio indicates general labour-market cooling.

3.3 Method

The analysis is descriptive. It reports peaks, averages over named windows, and ratios, and it compares the timing of turning points. No identifying assumption is required beyond the accuracy of the series, and no causal parameter is estimated. Where the data cannot distinguish two explanations, the paper says so and names the data that would.

4. Results

4.1 The entry door closed in October 2021

Information-sector hires peaked at 140,000 a month in October 2021. They were 94,000 in September 2026, a decline of 32.9 per cent, and the 2026 monthly average is 80,000. The first negative year-over-year reading in hires is February 2022: 18.3 per cent below the same month a year earlier. ChatGPT was released in November 2022, nine months later.

Figure 1 shows hires and employment on the same index. The hires line peaks and rolls over a year before the employment line does. That is the flow-stock relationship in one picture: employment is the integral of net hires, so it keeps rising for a year after hiring turns, and then it falls.

The hires rate tells the same story in a way that controls for the size of the sector. It averaged 3.11 per cent of employment in 2016–2019, peaked at 4.77 per cent in October 2021, and averages 2.83 per cent in 2026. The entry flow is now smaller relative to the sector than it was before the pandemic.

Figure 2 plots the hires rate. The peak in the fourth quarter of 2021 is the clearest turning point in the series, and it coincides with the peak in quits, which is the standard signature of a tight labour market rather than of a technology shock.

4.2 The contraction ran through attrition, not dismissal

Layoffs are the least important flow in this contraction. They averaged 33,000 a month in 2016–2019 and 49,000 in 2026, an increase of 50 per cent but a level far below the 191,000 reached in March 2020. The sector has not had a mass-separation event since the pandemic.

Quits fell instead. They averaged 44,000 a month in 2016–2019, peaked at 61,000 in October 2021, and average 34,000 in 2026, 23 per cent below the pre-pandemic norm. Workers stopped moving because the destination jobs disappeared, which is what a hiring freeze looks like from the inside.

The net position is the arithmetic that matters. Hires exceeded separations by an average of 11,200 a month in 2022. In 2026 the sector has averaged a net loss of 9,900 a month, and the hire-to-separation ratio has averaged 0.89. **Figure 3** shows the monthly net series: positive through 2022, negative and persistent from 2023 onward.

4.3 The arithmetic of a slow decline

Information employment peaked at 3.115 million in November 2022 and stands at 2.734 million in October 2026, a loss of 381,000 jobs, or 12.2 per cent, at an average of 8,100 a month over 47 months. That is a slow decline by the standards of a recession and a fast one by the standards of a hiring freeze, and it is exactly what a net flow of minus 8,000 to minus 10,000 a month produces.

The comparison with a layoff-led contraction is instructive. In March and April 2020 the sector lost 240,000 workers in a single month of separations, then rehired them within a year. The current decline has no such event: it is the cumulative effect of four years in which the sector hired fewer people than it lost.

4.4 A negative result on the entry-level hypothesis

If generative AI substitutes for junior work, the freeze should fall hardest on young workers. The aggregate cohort data give a mixed answer.

From the April 2023 trough to October 2026, unemployment rose 2.0 percentage points for ages 20–24 (5.5 to 7.5), 4.5 points for ages 16–19 (9.4 to 13.9), and 0.7 points for ages 25–54 (2.9 to 3.6). In levels, the young bore much more of the increase, which is consistent with the hypothesis.

The relative measure says otherwise. The ratio of the 20–24 unemployment rate to the 25–54 rate was 2.07 on average in 2016–2019, 2.12 in 2023, and 2.08 in October 2026. A shock that specifically reduced demand for entry-level work should raise that ratio. It has not moved outside its normal range.

Figure 4 plots the three unemployment rates. The young series are more volatile and rise more in a cooling labour market; the shape is familiar from every cycle in the sample. What the data show is a general cooling with the usual age gradient, not a new seniority-specific shock.

This is a negative result for the crude form of P1, and it is worth stating plainly because the crude form is what gets cited. It does not refute the hypothesis. It shows that aggregate cohort unemployment is too blunt an instrument: a hiring freeze that cuts the entry-level share of new hires in half would barely move the unemployment rate of 20–24 year-olds, because most of that cohort is not in the exposed sector, and because the composition of new hires is invisible in a rate that pools all employment statuses.

The test that would work is the entry-level share of new hires in exposed occupations, which requires job-posting composition or linked employer-employee data. Neither is in this paper.

4.5 Is the freeze distinctive to information?

No, and the paper should say so. Total nonfarm hires fell from 6.911 million a month in November 2021 to 5.402 million in September 2026, a decline of 21.8 per cent, and hires fell in professional and business services and in health care as well. The labour market cooled everywhere as the policy rate rose.

What is distinctive is the net position. Professional and business services hired less but continued to grow employment; health care continued to grow through the entire period. The information sector is the one where hires fell below separations and stayed there. **Figure 1** and the net series carry that point: a cooling labour market produces fewer hires everywhere; only in information did fewer hires produce a shrinking stock.

The information share of total hires fell from 2.10 per cent in October 2021 to 1.74 per cent in September 2026, a 17 per cent relative decline. That is the cleanest single statement of the sector's changing weight in the labour market, and it does not depend on any index base or counterfactual.

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5. What this identifies, and what it does not

The flow data narrow the mechanism. Any explanation of the information sector's contraction now has to run through the entry door: fewer hires, lower quits, modest layoffs, negative net flows. A story based on mass dismissal is inconsistent with the data. A story based on attrition and a hiring freeze is consistent with it.

The data do not identify the cause of the freeze. Three candidate causes all predict the same flows:

1. **Substitution.** Firms expect AI to do junior work and stop hiring for it.
2. **Cost of capital.** Firms facing higher discount rates cut discretionary expansion, and hiring is the first discretionary cost.
3. **Post-pandemic normalisation.** The sector over-hired in 2020–2021 and returned to trend.

The timing evidence favours 2 and 3 over 1. Hires turn in October 2021; the first negative year-over-year reading is February 2022; ChatGPT is released in November 2022. A technology that did not exist in public form cannot have closed the entry door nine months before it existed.

That argument is not decisive. Firms can anticipate a technology, and the research literature on language models was public before the product. But an anticipation story has to explain why the freeze is concentrated in the exact quarter when the rate cycle turned, and why the sector's hiring did not accelerate again in 2023–2024 as model capability improved.

The composition data would be decisive in a way that timing is not. If the freeze is substitution, the entry-level share of hires should fall even while total hires recover. If it is cost of capital, the freeze should be broad across seniority and concentrated in firms with weak balance sheets. No public data set currently separates those two predictions for this sector.

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6. What to measure

The paper's constructive output is a short list, ordered by how much it would resolve.

Hiring composition by seniority. The entry-level share of new hires in exposed occupations, quarterly. This is the single statistic that would distinguish substitution from a broad freeze, and it is visible in job-posting data and in linked employer-employee records.

Separation type by tenure. Attrition versus dismissal, split by tenure. A substitution story predicts junior attrition that is not replaced; a cost story predicts broad-based reduction in postings.

Firm-level exposure and hiring. Postings and headcount by firm, joined to a measure of AI adoption (spend, usage, or tool deployment). Exposure scores are not a substitute; the companion paper argues that adoption is the missing variable.

Reinstatement. Hires in the same occupations two years later. The task framework predicts that displaced tasks are eventually reinstated in new combinations; a freeze that never thaws is evidence against the framework's optimistic branch.

None of these require a new exposure index. All of them require administrative or commercial microdata that the public JOLTS does not contain.

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7. Threats to validity

Flow measurement. JOLTS counts hires and separations as events, not unique workers, and the survey is subject to revision. The most recent months are provisional. Seasonal adjustment can move turning points by a month or two, which matters for an argument that leans on October 2021 versus November 2022.

Sector definition. Information is a narrow establishment-based sector: software publishers, telecommunications, media, data processing. A software engineer at a bank is counted in finance. The sector's contraction therefore measures a subset of technology employment, and the true technology labour market is larger and differently distributed.

Timing inference. The argument that the freeze predates ChatGPT rests on the 2021 peak and the February 2022 year-over-year turn. Firms can anticipate technology, and the peak in the hires rate coincides with the tightest labour market in decades, which makes the 2021 value an outlier in the other direction. A reader who rejects the timing inference still has the mechanism result, which does not depend on dates.

Cohort aggregation. The entry-level test uses unemployment rates for broad age bands. Most 20–24 year-olds are not in AI-exposed occupations; the rate pools them. A null result on the ratio is therefore weak evidence against the hypothesis, not strong evidence for it.

Single sector. The paper studies one sector. The mechanism may generalise; the magnitude does not.

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8. Reproducibility

```
data/series.csv          # 130 monthly rows, 30 series, BLS
analysis/fetch.py        # pulls the series from the BLS API, caches raw JSON
analysis/analyze.py      # every number in this paper, stdlib only
analysis/stats.json      # the numbers, machine-readable
figures/fig1..4.svg      # regenerated by analyze.py
```

```
python3 analysis/fetch.py      # requires network; writes data/
python3 analysis/analyze.py    # offline; writes stats.json and figures
```

Every number in Sections 4 and 5 traces to a field in `analysis/stats.json`. Two checks a reader can run: `hires.peak.date` is 2021-10-01 and `net_hires.avg_2026` is −9.9.

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9. Conclusion

The information sector lost 381,000 jobs in 47 months without a layoff wave. It hired less, lost fewer workers to quits, and let the arithmetic do the work: hires below separations, month after month, at about minus 8,000.

The entry door closed in October 2021, thirteen months before employment peaked and before the technology that is supposed to have caused the contraction was public. That does not prove the technology is innocent. It does establish the order of events, and it narrows the mechanism: whatever happened to this sector ran through hiring, not firing.

The entry-level hypothesis fails its crudest test. Youth unemployment rose more than prime-age unemployment in percentage points, but the ratio between them stayed inside its normal range, which is what a general cooling looks like. The composition change the hypothesis predicts is invisible at the level of a cohort unemployment rate.

Two conclusions follow. For research: the next four years of AI-and-labour work should be measured at the entry door, with hiring composition by seniority, not at the exit door with layoff counts, and not at the level of exposure scores that never observe a hire. For policy: a labour market adjusting through a hiring freeze is invisible in the unemployment rate for a long time and very visible to the people who cannot get their first job. The statistic that catches it is not the one currently in the headline.

Appendix A. Series and coverage

Series ID	Name	First	Last	Missing months
JTS5100000000000000HIL	Information hires	2016-01	2026-09	0
JTS5100000000000000TSL	Information separations	2016-01	2026-09	0
JTS5100000000000000LDL	Information layoffs	2016-01	2026-09	0
JTS5100000000000000QUL	Information quits	2016-01	2026-09	0
JTS5100000000000000JOL	Information openings	2016-01	2026-09	0
CES5000000001	Information employment	2016-01	2026-10	0
LNS14000012	Unemployment 16–19	2016-01	2026-10	0
LNS14000036	Unemployment 20–24	2016-01	2026-10	0
LNS14000060	Unemployment 25–54	2016-01	2026-10	0

Appendix B. Figures

Figure 1. Information sector hires and employment, index January 2019 = 100.

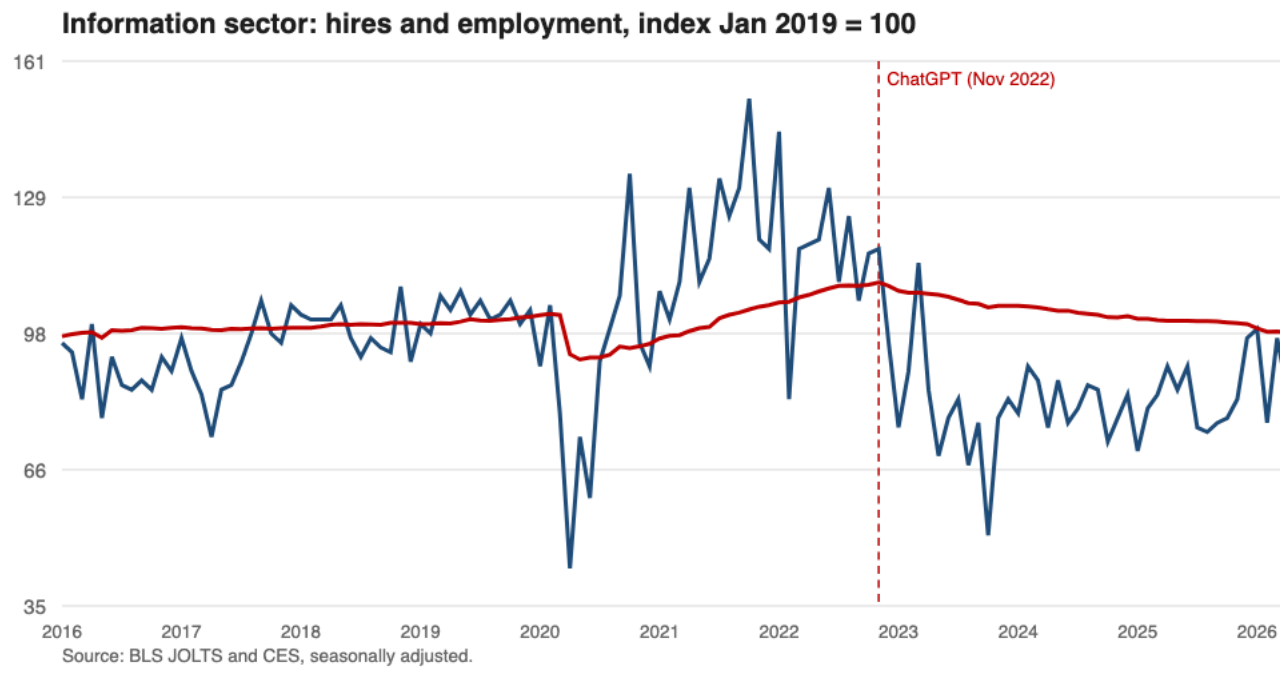


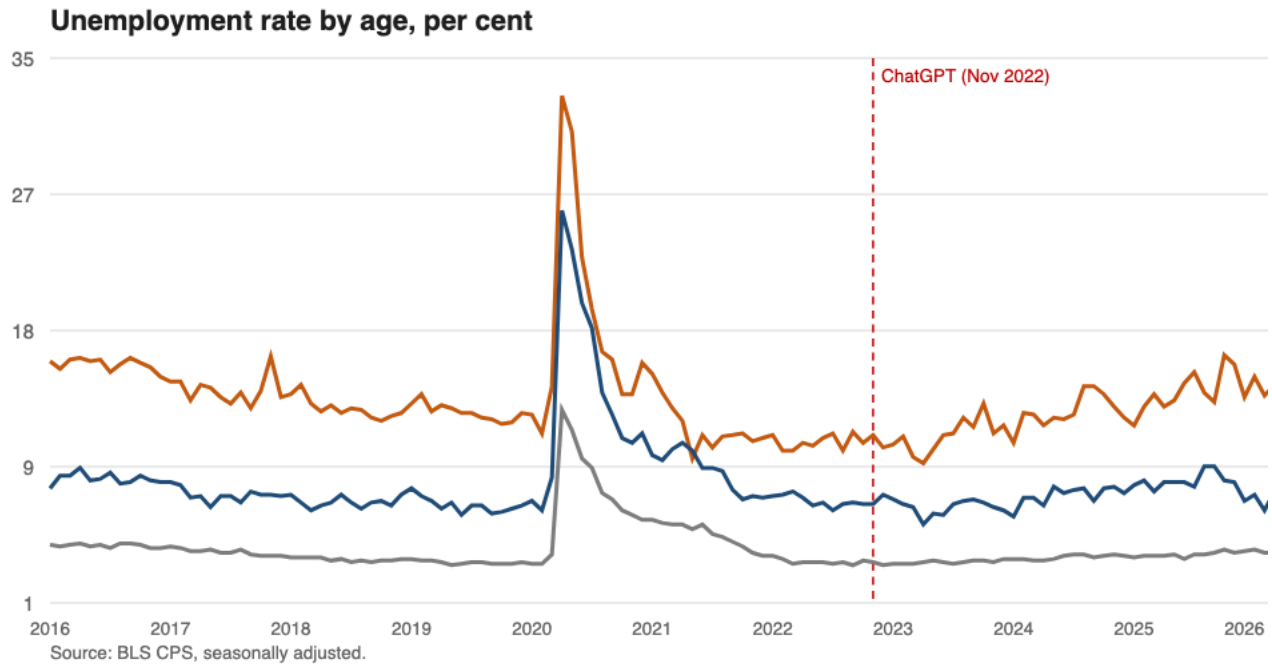
Figure 2. Information sector hires rate, hires as a percentage of employment.



Figure 3. Information sector monthly net hires, hires minus separations, thousands.



Figure 4. Unemployment rate by age cohort, per cent.



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